

EGCO 425 – Chapter 2

Data Exploration

Data Visualization

References

- [1] Kotu & Deshpande, Data Science: Concepts and Practice (Chapter 3)
- [2] Tan et al., Introduction to Data Mining (Chapters 2-3)
- [3] RapidMiner Doc, <https://docs.rapidminer.com/latest/studio/operators/>

Objectives of Data Exploration

- ⊕ To understand the data
 - ▶ Types of attributes, number of attributes, number of records
 - ▶ Typical value of each attribute
 - ▶ Potential problems such as errors, missing values, outliers
- ⊕ To prepare the data (before applying advanced analysis)
- ⊕ As part of data analysis, e.g.
 - ▶ Identify relationship between attributes
 - ▶ Detect clusters of data
- ⊕ To interpret the results

Measurement Scales of Attributes

- ⊕ In statistics and data mining, scales or types of attributes depend on the properties they possess

Property		Type		
D istinctness	= ≠	Nominal	possess	D
O rder	> <	Ordinal		D O
A ddition	+ −	Interval		D O A
M ultiplication	* /	Ratio		D O A M

Qualitative (Categorical) Attributes

⊕ **Nominal**

- ▶ Values are names or categories
- ▶ Provide just enough information to **distinguish** one object from another
- ▶ Statistics : mode, entropy, correlation
- ▶ Permissible transformation is simply permutation of values, e.g.
 - {red, green, blue} → {green, blue, red}
order of values is not important
- ▶ In RapidMiner's term, nominal attribute has 2 subtypes
 - Binominal only 2 possible values = binary in some text books
values can be bits (0/1) or boolean (false/true)
 - Polynominal >2 possible values

⌘ Ordinal

- ▶ Provide enough information to **order** objects
- ▶ But magnitude between successive values is not known/not constant
 - Size S (waist = 26 in) and size M (waist = 28 in) → diff = 2 in
 - Size M (waist = 28 in) and size L (waist = 31 in) → diff = 3 in
- ▶ Statistics : median, percentile
- ▶ Permissible transformation must preserve order, e.g.
 - Grade {A, B, C, D} → grade {4, 3, 2, 1}
Change values, but keep order info. (A is mapped to 4)

Quantitative (Numeric) Attributes

Interval

- ▶ The difference between values is meaningful (but ratio is not); equal magnitude between successive values; no "true zero"
 - Calendar dates, size in number, IQ score
 - Shirt size → between 10 and 12 diff = 2
 between 14 and 16 diff = 2
 - Temperature in °C or °F → 0°C ≠ no heat
 2°C ≠ twice the heat of 1°C
 - Relative height → Bill's height = 2 inches above average
 Bob's height = 4 inches above average
 Bob ≠ twice as tall as Bill
- ▶ Statistics : mean, stdev
- ▶ Permissible transformation takes scale and unit into account
 - E.g. °C and °F have different scales, so $C/5 = (F-32)/9$

⊕ Ratio

- ▶ Both **difference** and **ratio** are meaningful; "true zero" exists
 - Temperature in Kelvin → 0 K (-273 °C) = no heat
2 K = twice the heat of 1 K
 - Age
 - Size in in/cm, length
 - Amount of money
- ▶ Statistics : mean, geometric mean, harmonic mean
- ▶ Permissible transformation : arithmetic operations

Descriptive Statistics

Characteristics of the Dataset	Measurement Technique
Center of the dataset	Mean, median, and mode
Spread of the dataset	Range, variance, and standard deviation
Shape of the distribution of the dataset	Symmetry, skewness, and kurtosis

⊕ Measure of central tendency

- ▶ Numeric attribute mean, median
- ▶ Nominal attribute mode

⊕ Measure of spread → for numeric attribute

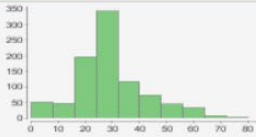
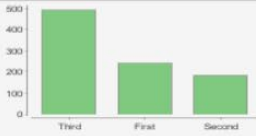
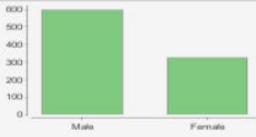
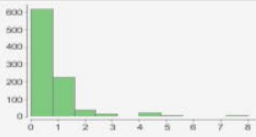
- ▶ Range = max value – min value

$$\text{Variance} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad ; \text{ standard deviation (SD)} = \sqrt{\text{variance}}$$

Use n-1 for sample, N for population

⊕ Shape of distribution

- ▶ Numeric attribute histogram, normal distribution curve
- ▶ Nominal attribute frequencies & relative frequencies

Age	Real	0		Min 0.167	Max 80	Average 29.932	Deviation 13.265
Passenger Class	Polynomial	0		Least Second (184)	Most Third (491)	Values Third (491), First (241), Second (184) Details...	
Sex	Binominal	0		Negative Female	Positive Male	Values Male (594), Female (322) Details...	
No of Siblings or Spouses on B...	Integer	0		Min 0	Max 8	Average 0.514	Deviation 1.014

Descriptive stat from RapidMiner

Data Visualization

⊕ Reasons

- ▶ Comprehension of dense information → can easily see the big picture and long-term trends
- ▶ Relationship between attributes

⊕ 2D Cartesian (X-Y) axis is recommended

- ▶ X-axis input, independent variable, nominal categories
- ▶ Y-axis output, dependent variable, numeric values

⊕ Use 3D only if the 3rd dimension carries additional information

- ▶ If 2D chart is sufficient, there's no need for 3D chart

Plot Types

⊕ Visualizing 1 attribute

- ▶ Histogram, boxplot, bell curve

distribution of numeric attributes

⊕ Visualizing 2-3 attributes

- ▶ Based on X-Y axis and colors
- ▶ Boxplot, bell curve
- ▶ Bar chart, line chart
- ▶ Pie/donut chart, sunburst chart
- ▶ Scatter plot, scatter matrix

distribution of numeric attributes

comparison of categories

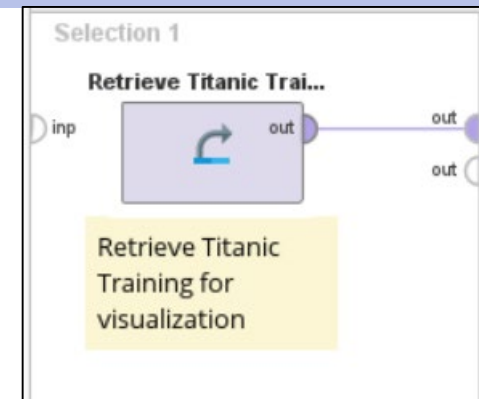
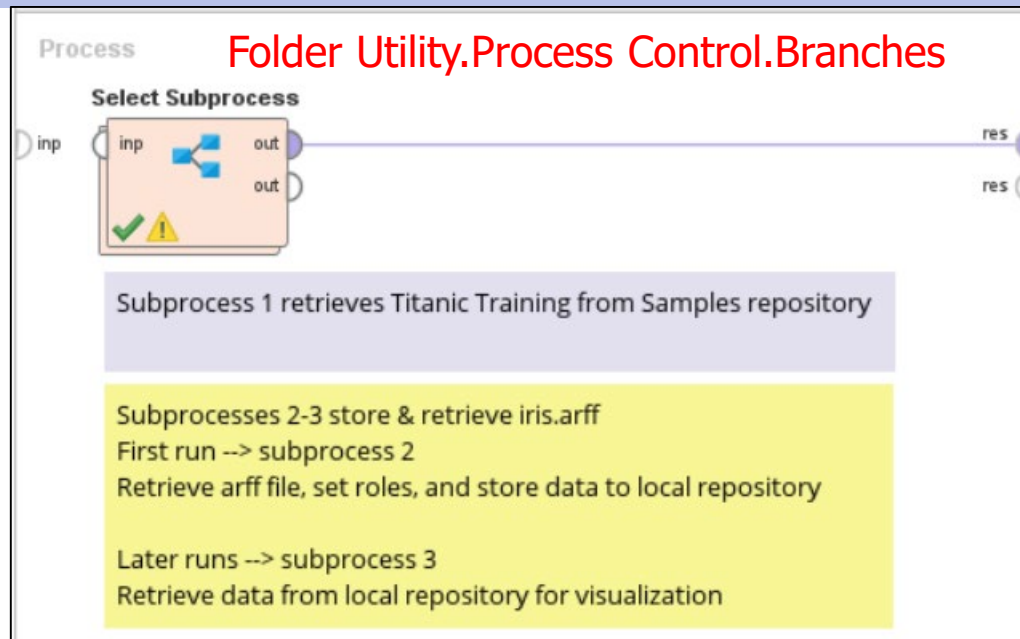
comparison of categories

relationship of numeric attributes

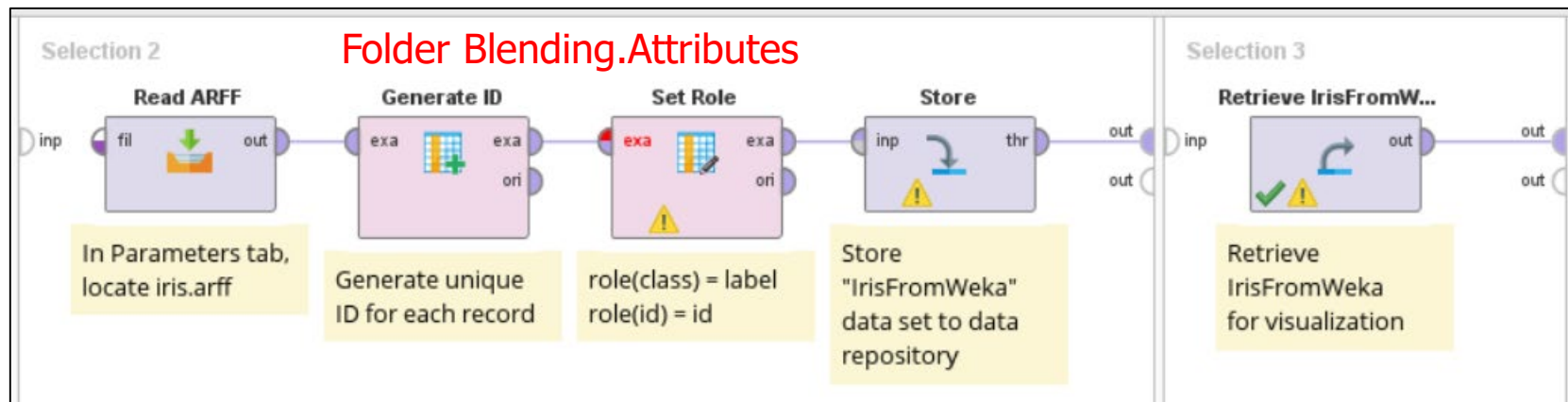
⊕ Visualizing >3 attributes

- ▶ An axis is shared by >1 attributes
- ▶ Parallel chart, deviation chart

Example (1) : retrieval and visualization



Or use Turbo Prep instead of Running subprocesses 1 & 3



Attribute Roles in RapidMiner

- ⊕ ID = unique ID for each record
- ⊕ Regular = input attribute = default role

Learning or classification task

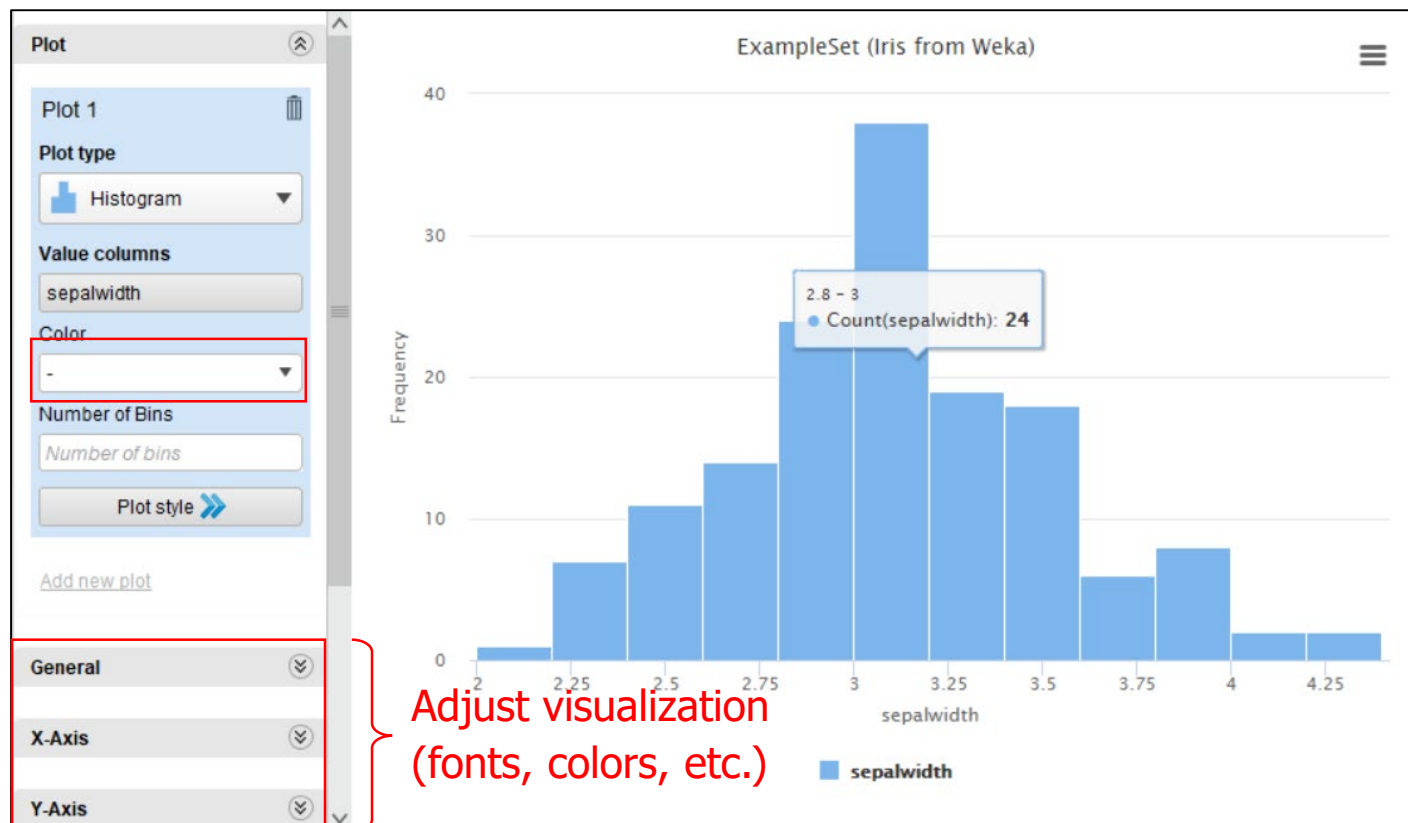
- ⊕ Label = target variable or class
- ⊕ Prediction = classification result (given by learning model)
- ⊕ Weight = weight of record w.r.t. label

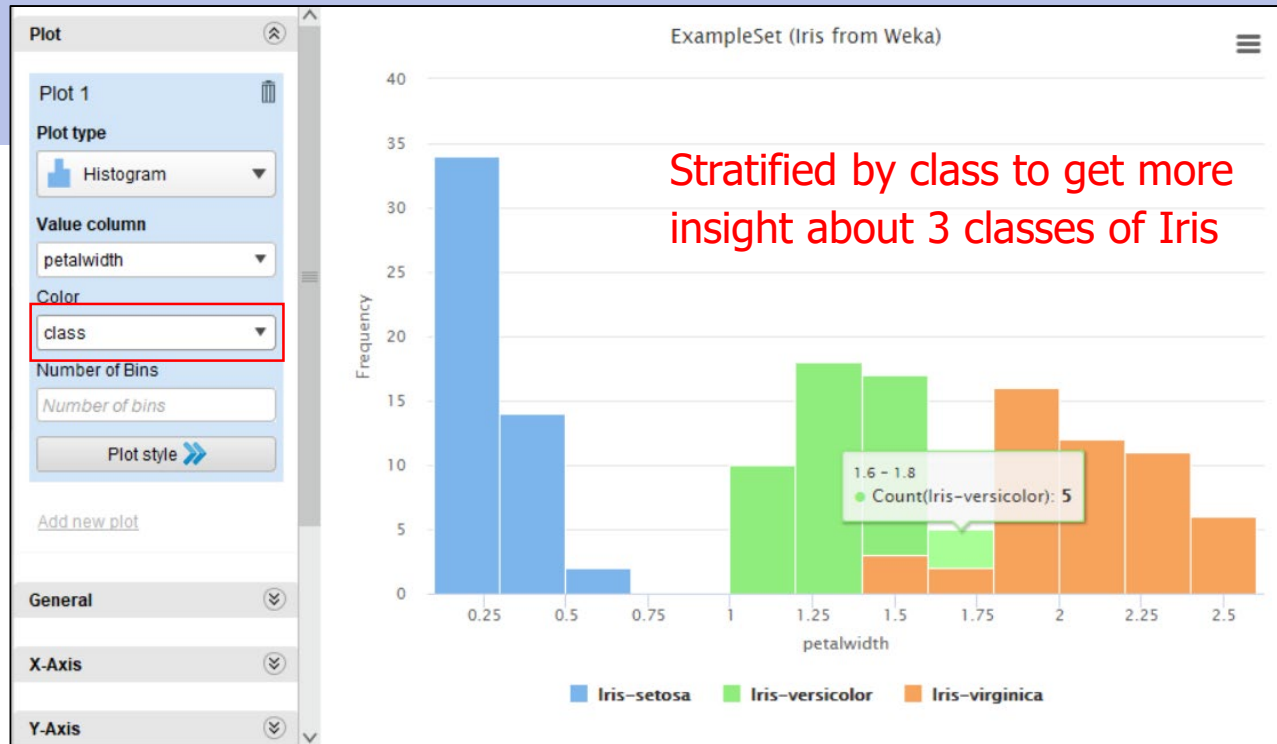
Clustering task

- ⊕ Cluster = cluster membership (assigned by clustering model)

Histogram

- Show data distribution by plotting the frequency of values (Y-axis) in each range or bin of numeric values (X-axis)





Petalwidth of Setosa is clearly smaller than Petalwidth of the others



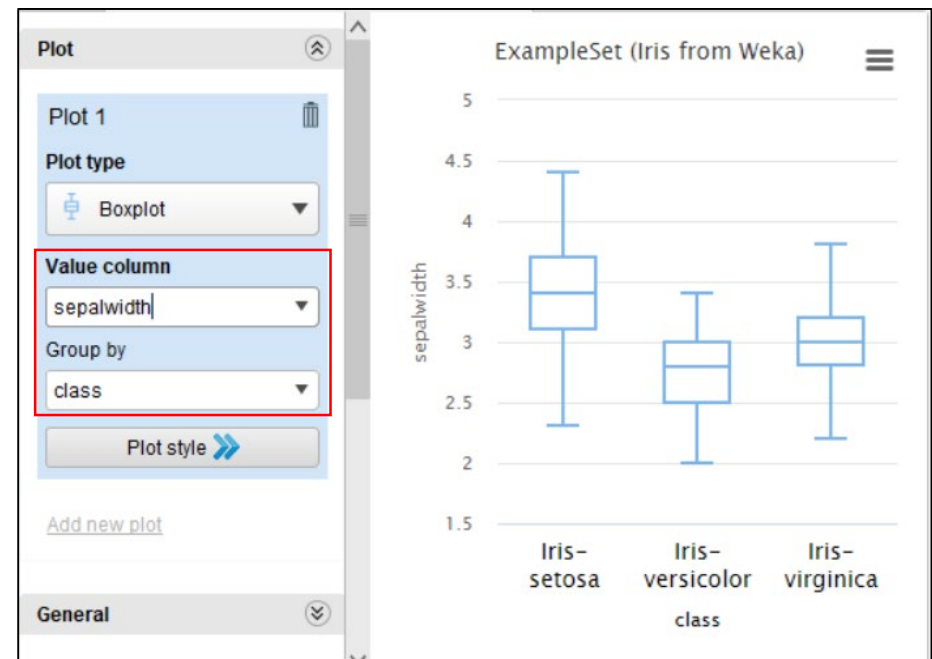
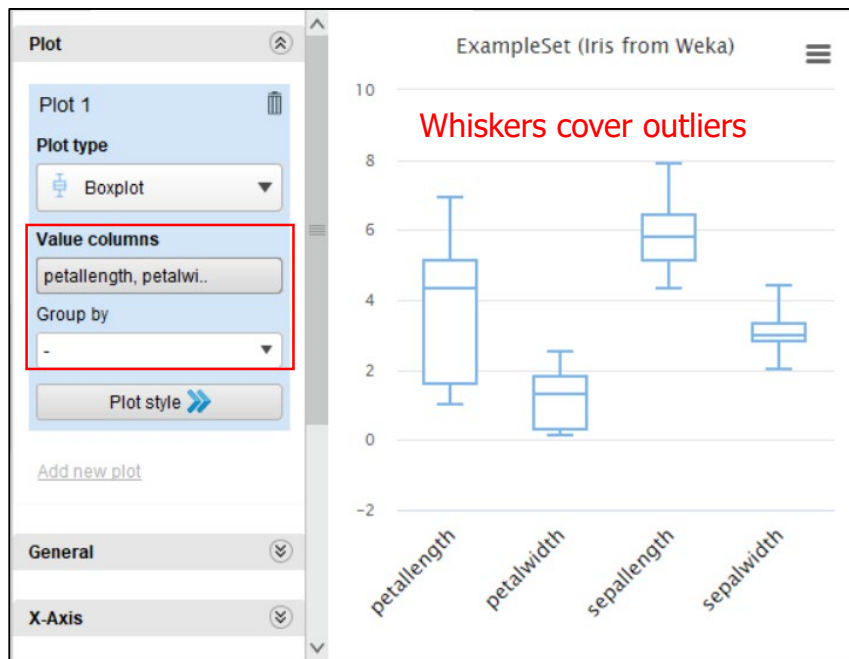
Setosa



Virginica

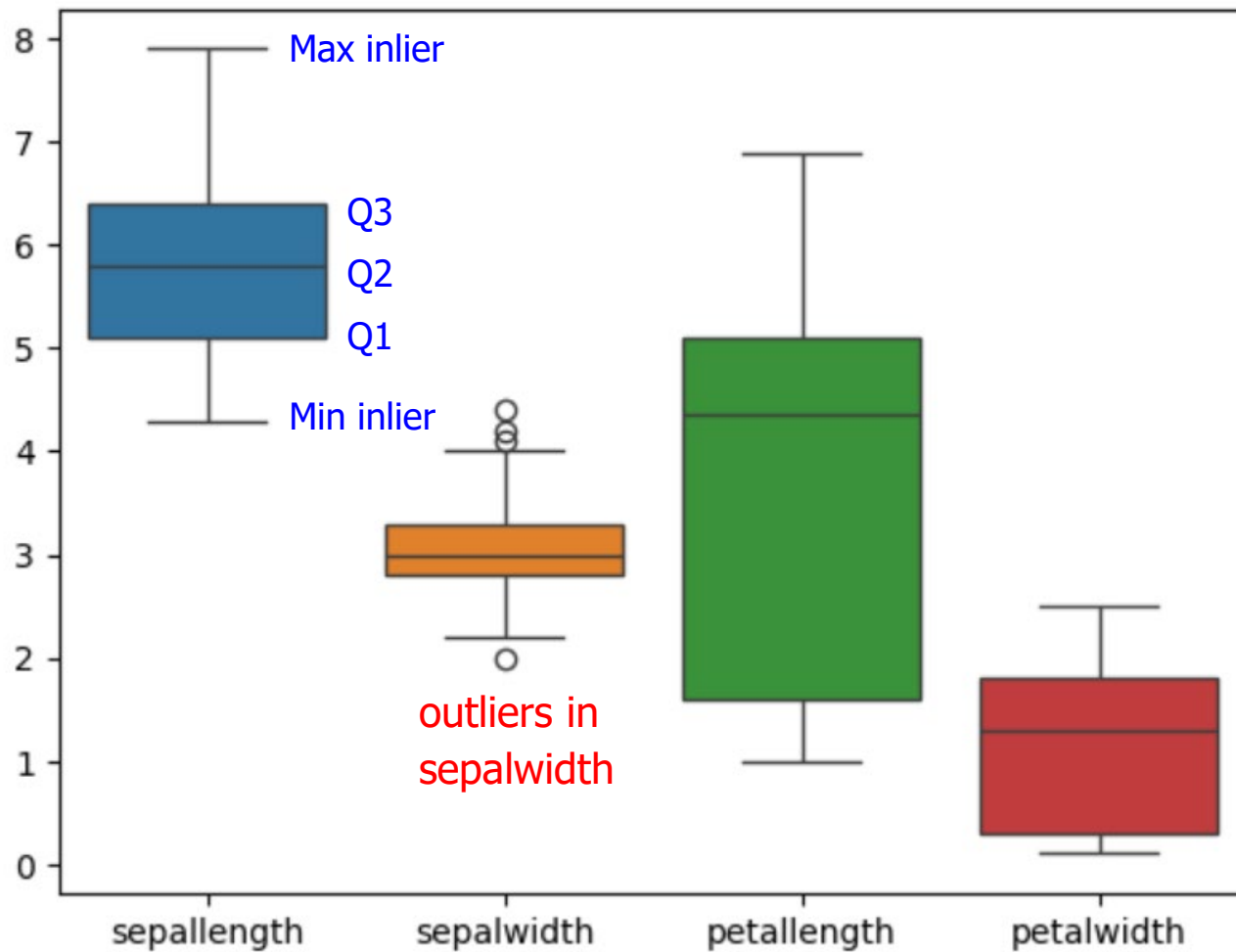
Boxplot

- ✦ Show data distribution along with quartiles, median, outliers
- ✦ Compare distribution of different **numeric attributes** or categories
- ✦ But RapidMiner doesn't show outliers in boxplot → use Python



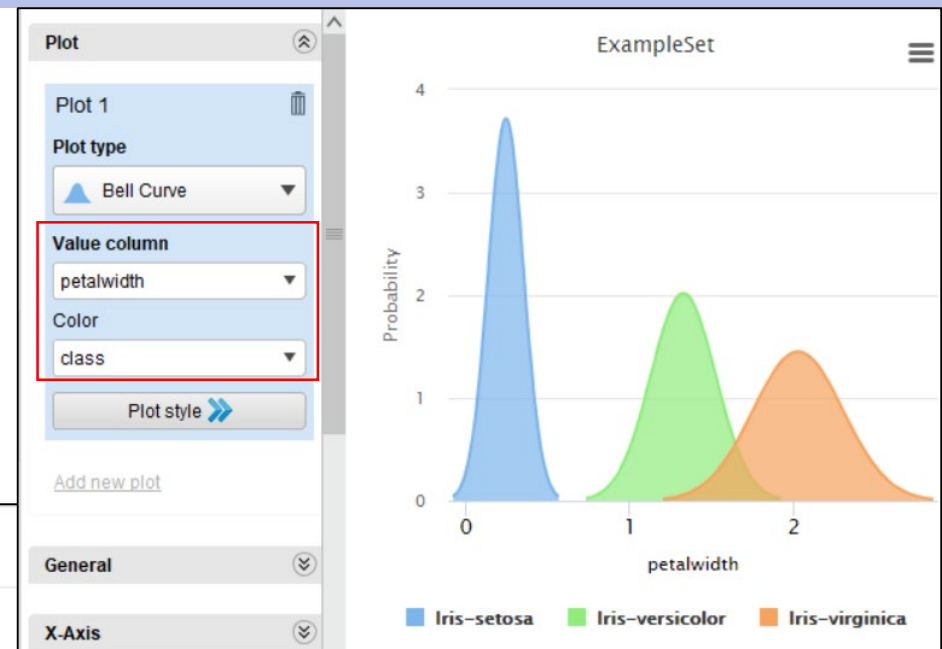
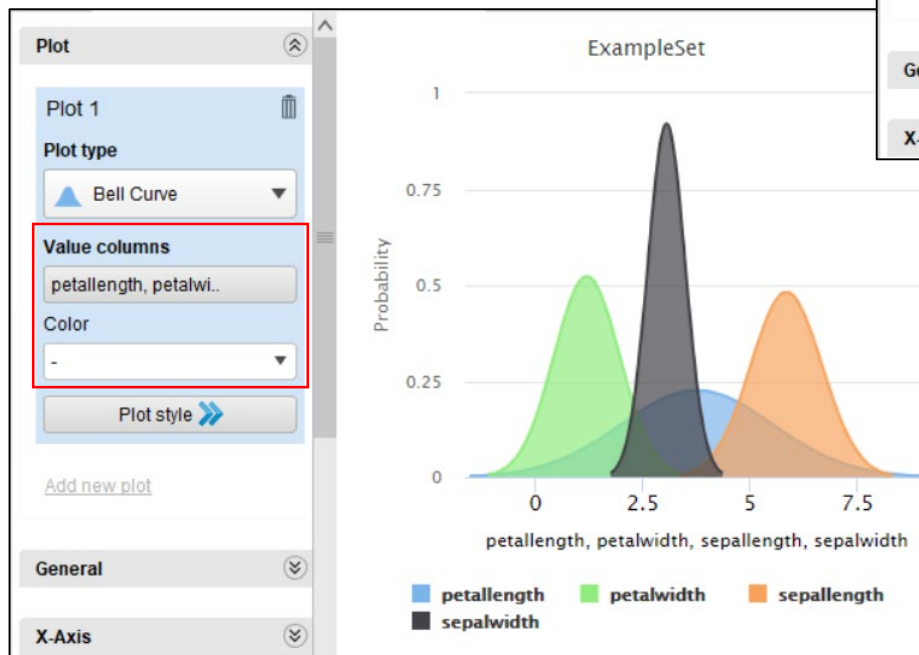
Sepalwidth has little variation, compared to petallength

Boxplot by Python



Bell Curve

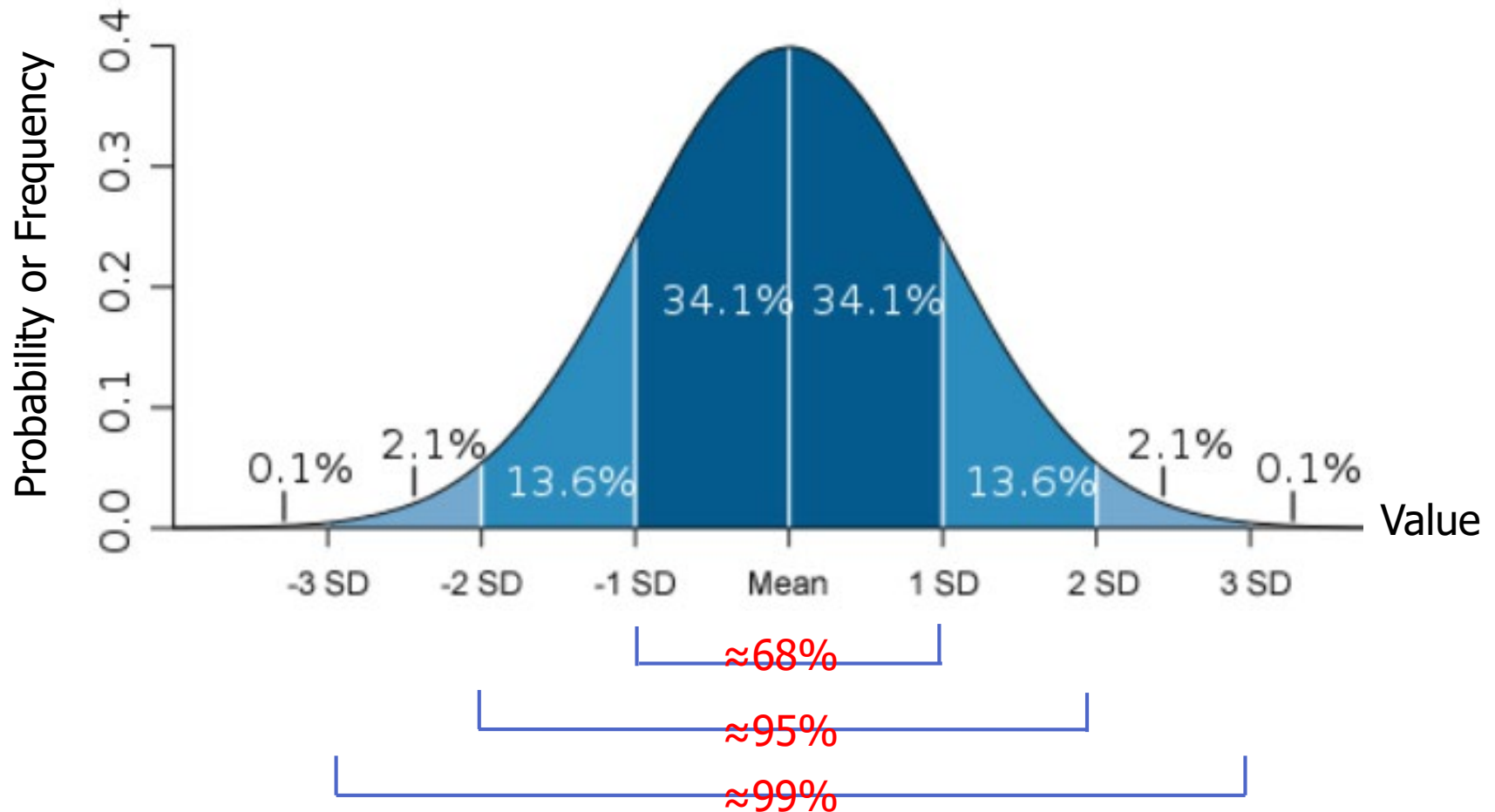
- Show normal probability distribution (mean & spread)
- Compare mean & spread in different **numeric attributes** or categories

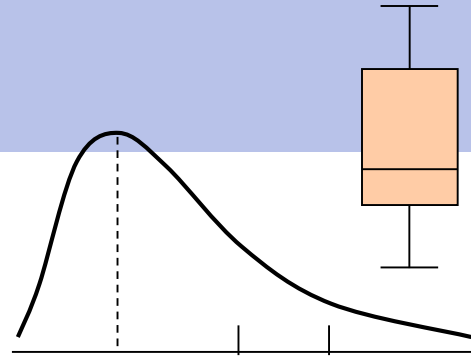


Unlike statistical methods, most data mining methods don't require normality assumption

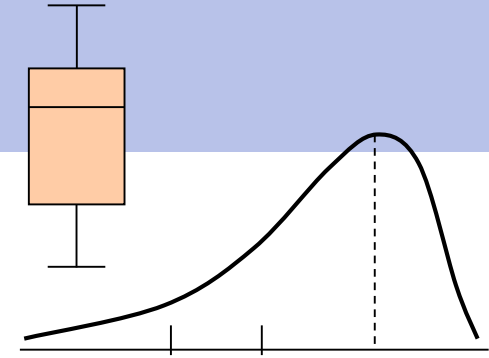
Instead, we use evaluation metrics to measure whether a model fits the data

Normal Distribution

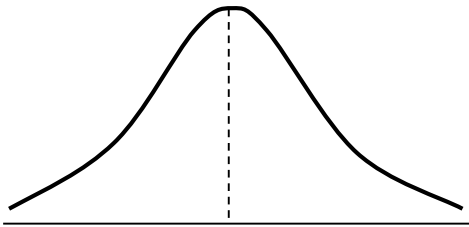




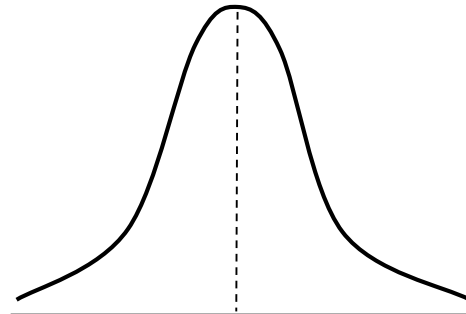
mode < median < mean
Positive (right) skewness
 -- long tail to the right



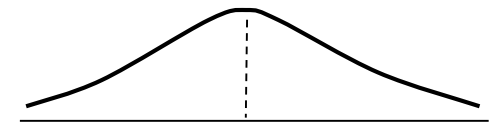
mean < median < mode
Negative (left) skewness
 -- long tail to the left



Normal distribution
 mean = mode = median
 Zero skewness
 Zero kurtosis



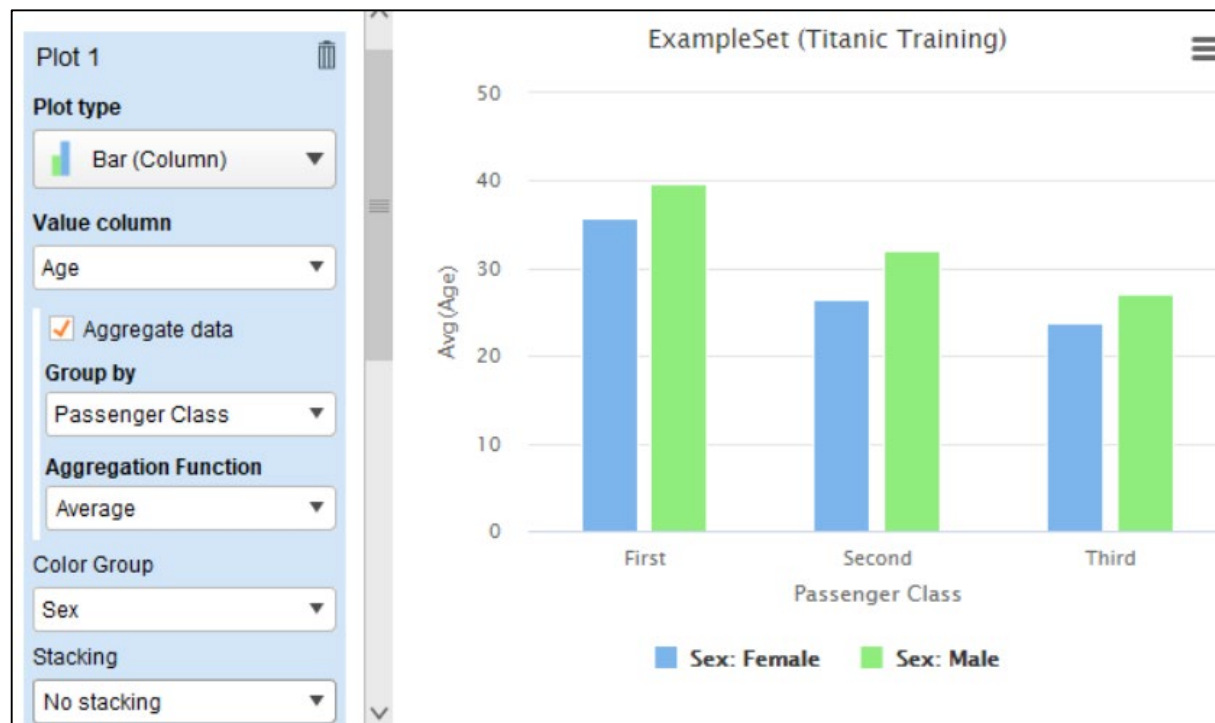
Positive kurtosis
 (high peak, narrow base)



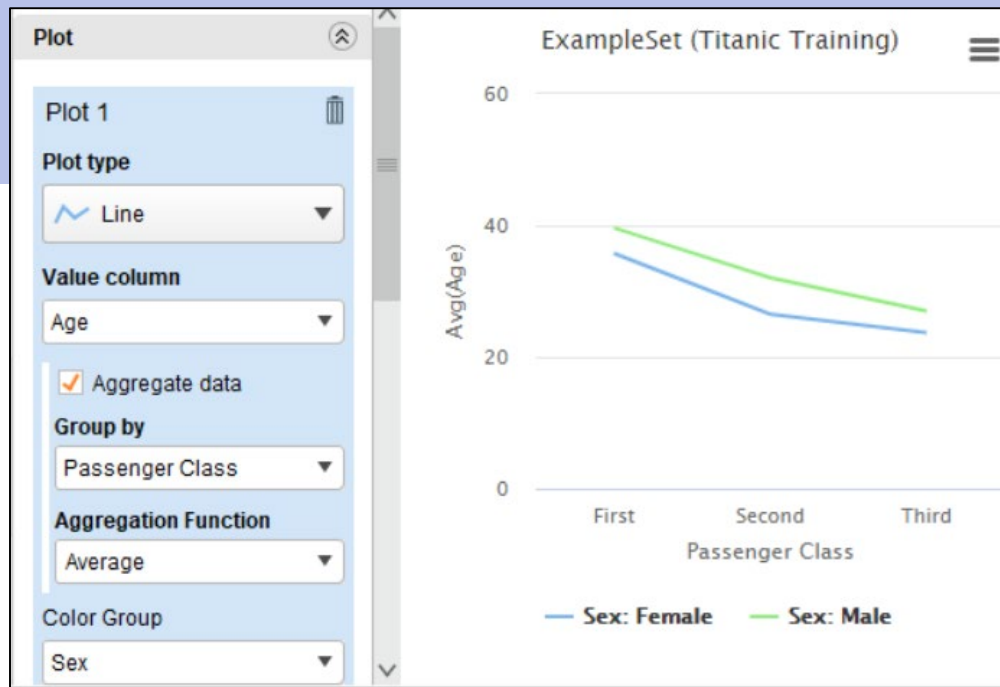
Negative kurtosis
 (low peak, wide base)

Bar Chart, Line Chart

- ✦ Compare **numeric values (Y-axis)** of different **categories of nominal attributes (X-axis and colors)**

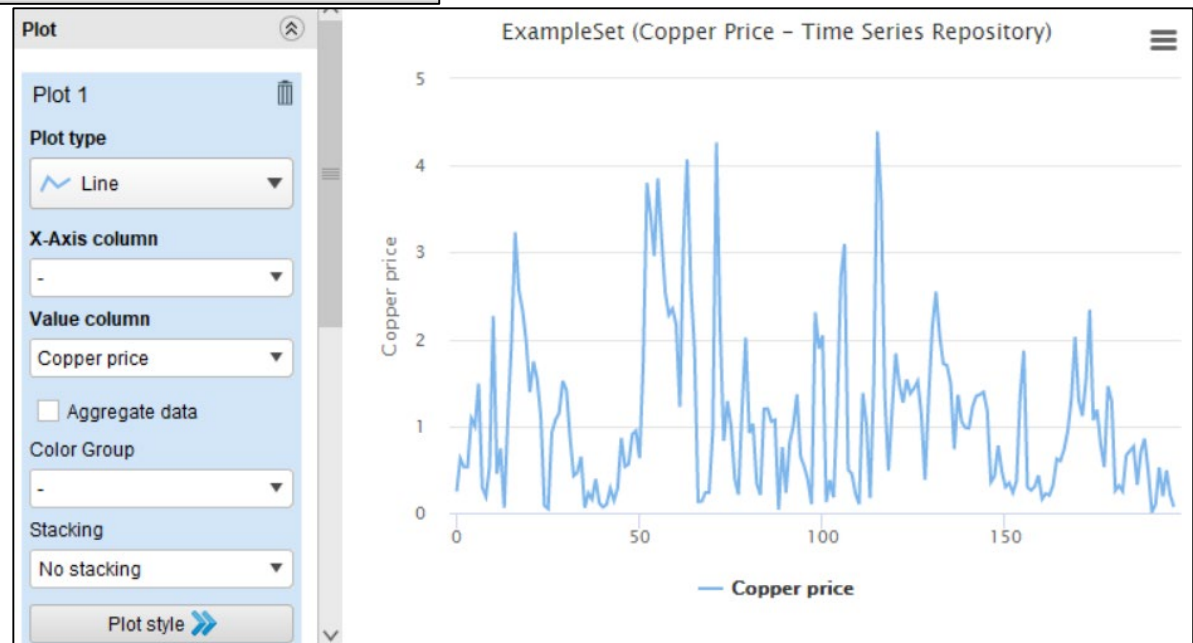


Average age of female and male passengers in different passenger classes



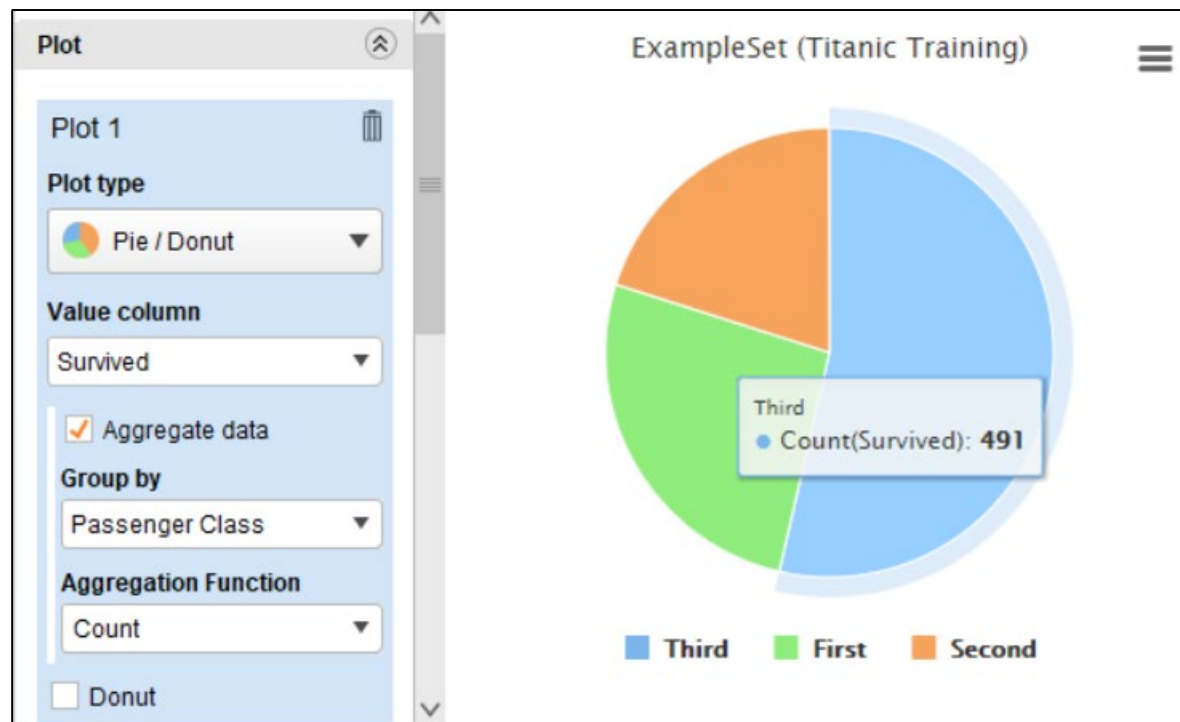
Line chart can be used as an alternative of bar chart

or to show trend of data over a period of time (X-axis)



Pie/Donut Chart

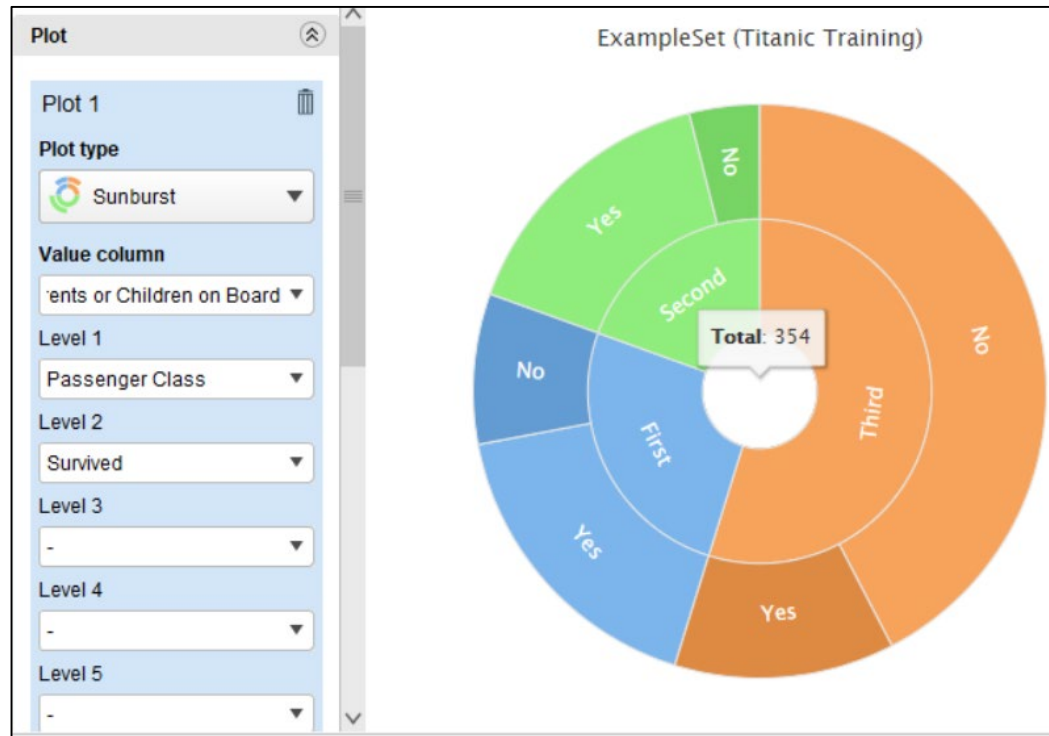
- ✦ Compare proportions or frequencies (of a nominal attribute) in different categories of another nominal attributes



Out of all survivors (in the whole data set),
how many of them are in different passenger classes

Sunburst Chart

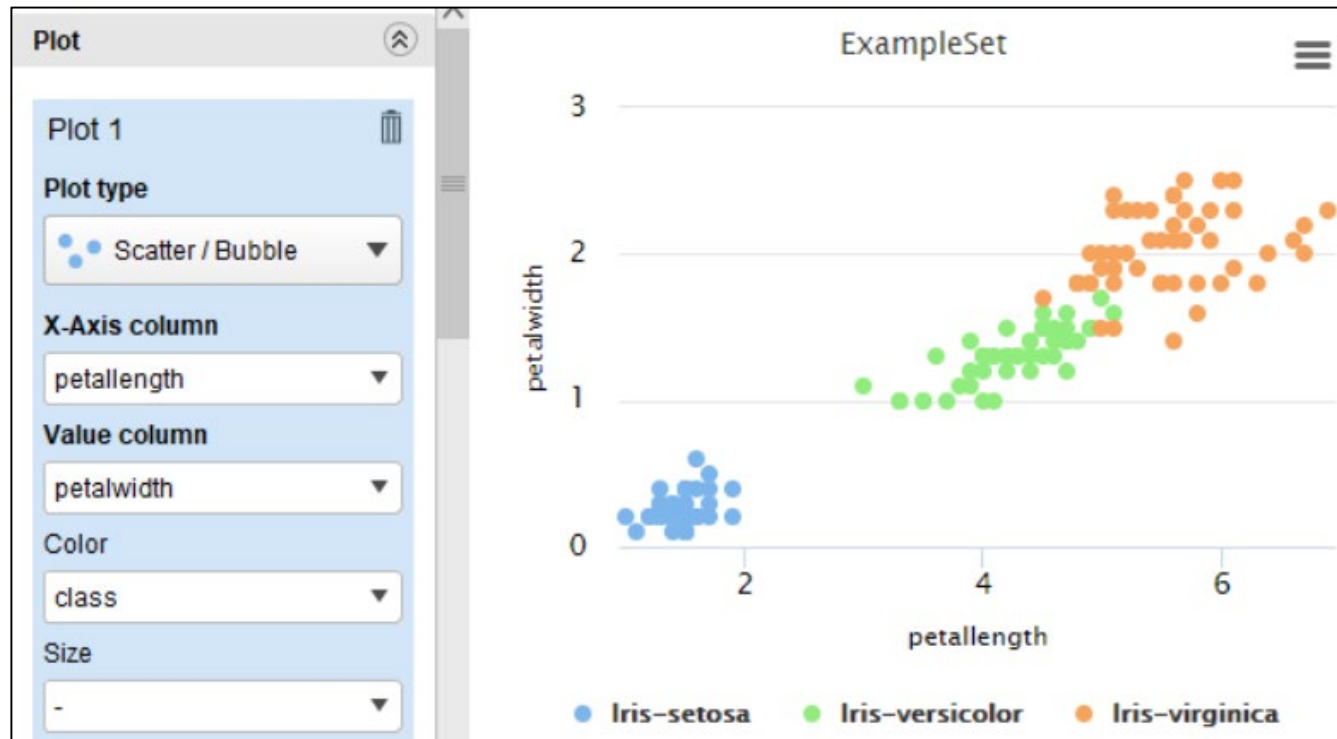
- ✦ Compare proportions in different categories of multiple nominal variables (in hierarchy)



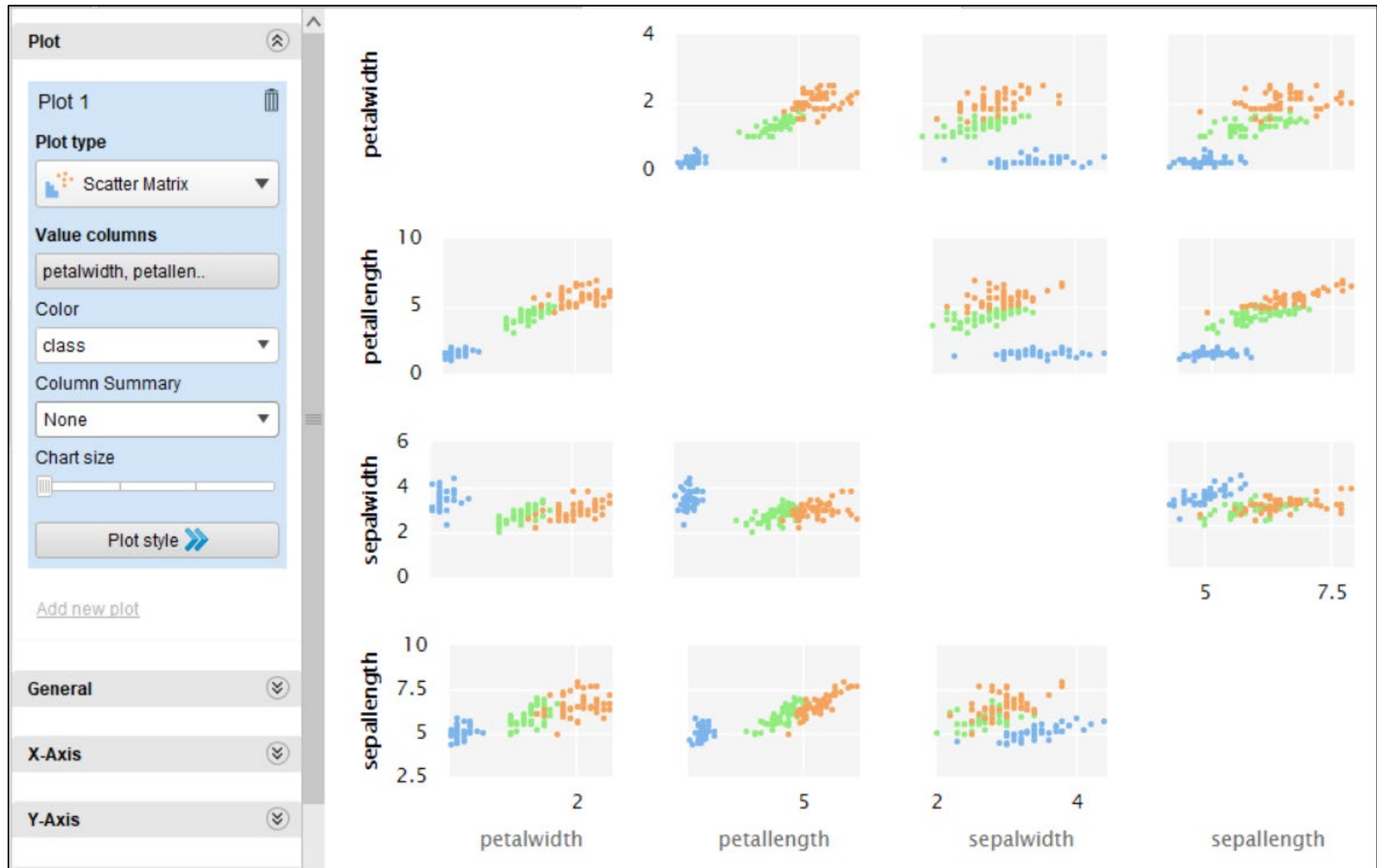
Out of 354 parent/children on board (in the whole data set), how many of them are in different categories of passenger class & survived

Scatter Plot

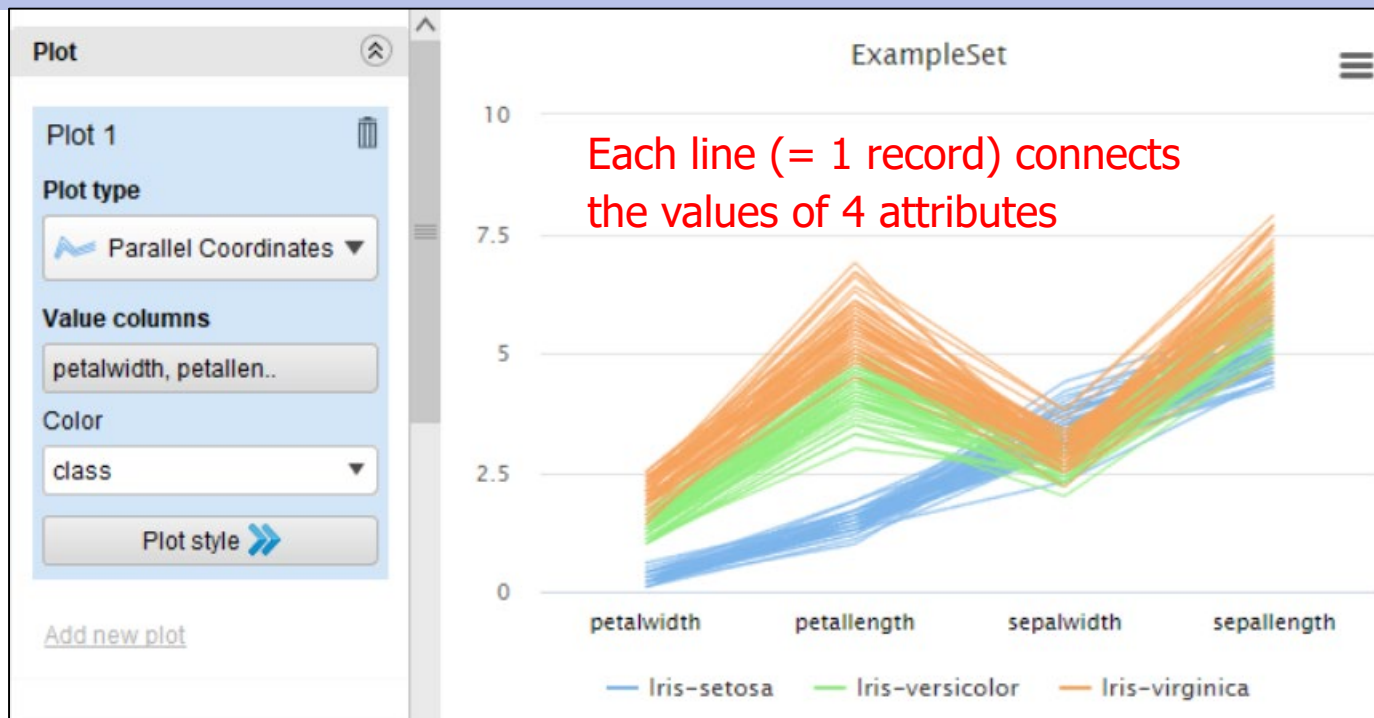
- ✦ Show relationship patterns between 2 numeric variables



Scatter Matrix (of multiple scatter plots)



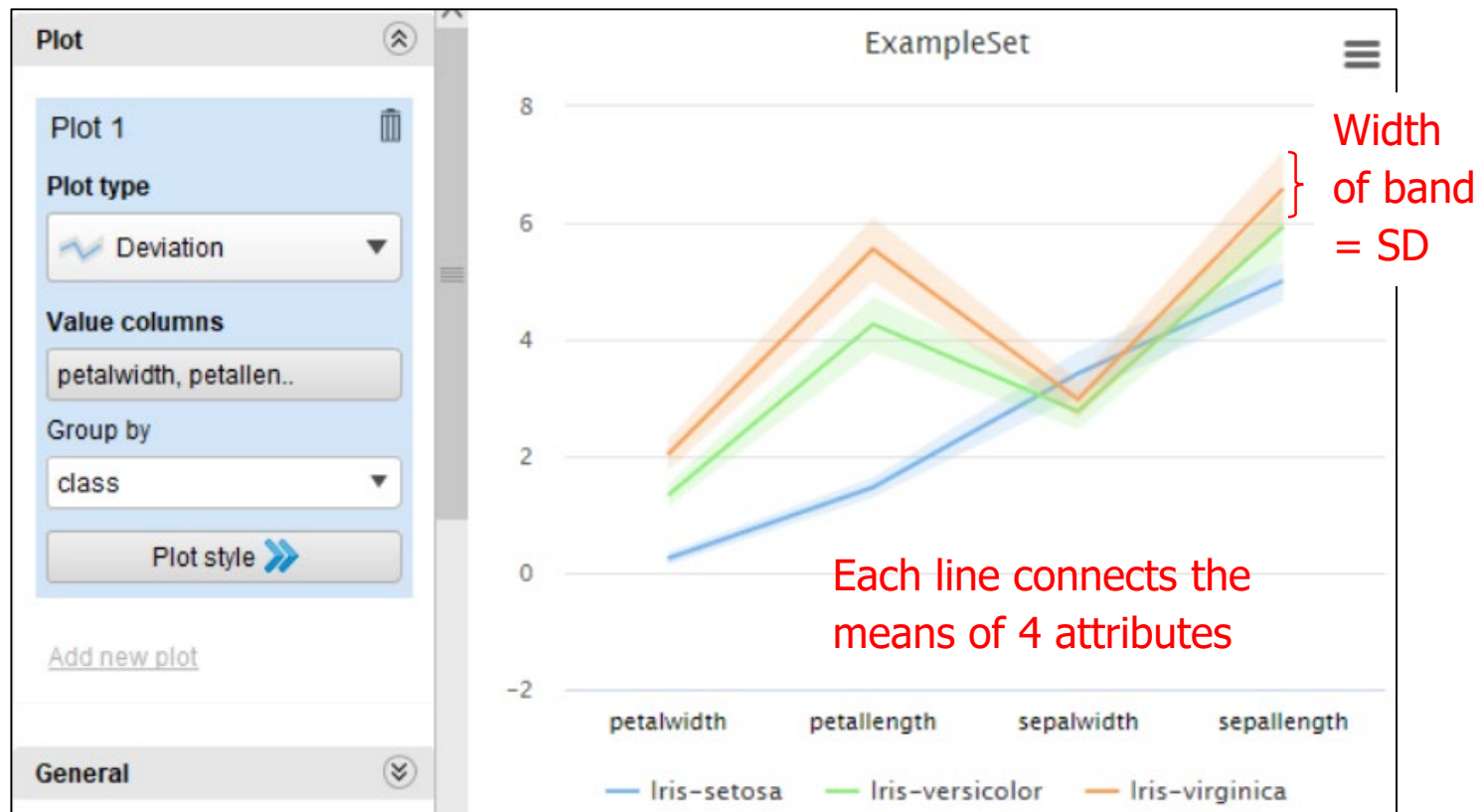
Parallel Chart



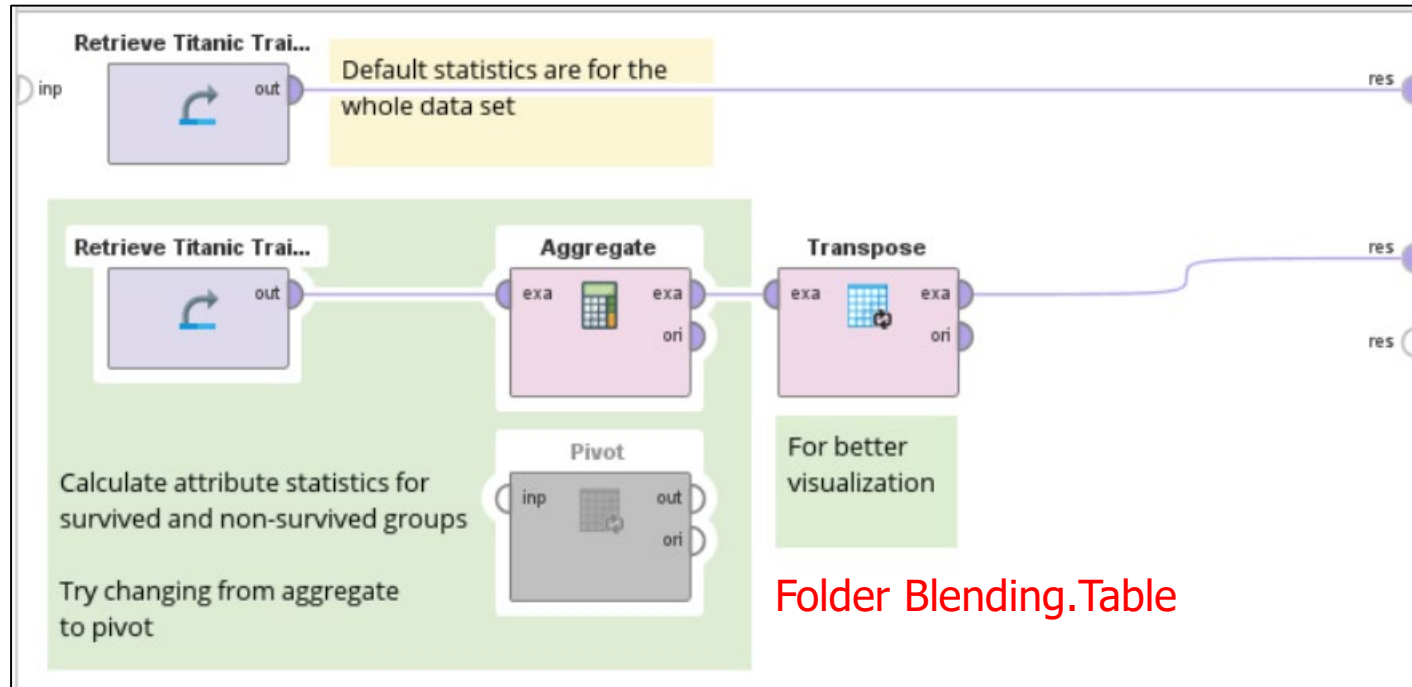
- ✦ Attributes are placed on X-axis
- ✦ Since Y-axis is shared by many attributes, attributes should be normalized to have similar range
- ✦ Useful for quick observation of class differences exhibited by each attribute

Deviation Chart

- ✦ Instead of showing individual lines for individual records (as in parallel chart) → only mean & SD of each category are shown
- ✦ Useful for quick assessment of the spread of each class for each attribute



Example (2): per-group statistics (aggregation & pivot)



Result from Aggregate

Row No.	id	att_1	att_2
1	count(Survived)	567.0	349.0
2	mode(Sex)	Male	Female
3	mode(Passenger Class)	Third	First
4	average(Age)	30.501825...	29.005654...
5	minimum(No of Parents or Children on Board)	0.0	0.0
6	maximum(No of Parents or Children on Board)	9.0	4.0
7	sum(No of Siblings or Spouses on Board)	300.0	171.0
8	average(Passenger Fare)	24.359743...	48.516822...
9	Survived	No	Yes

Attribute statistics grouped by survived

Result from Pivot

Row No.	id	att_1	att_2
1	Survived	Yes	No
2	count(Survived)_Female	239.0	83.0
3	count(Survived)_Male	110.0	484.0
4	mode(Passenger Class)_Female	First	Third
5	mode(Passenger Class)_Male	Third	Third
6	average(Age)_Female	28.945582...	25.609616...
7	average(Age)_Male	29.136175...	31.340779...
8	minimum(No of Parents or Children on Board)_Female	0.0	0.0
9	minimum(No of Parents or Children on Board)_Male	0.0	0.0
10	maximum(No of Parents or Children on Board)_Female	4.0	6.0
11	maximum(No of Parents or Children on Board)_Male	2.0	9.0
12	sum(No of Siblings or Spouses on Board)_Female	122.0	78.0
13	sum(No of Siblings or Spouses on Board)_Male	49.0	222.0
14	average(Passenger Fare)_Female	53.281939...	22.984440...
15	average(Passenger Fare)_Male	38.163523...	24.595591...

Attribute statistics grouped by survived and sex

Roadmap for data exploration

1. Organize data set to have records in rows & attributes in columns
 - Check size of data set (#records, #attributes, types of attributes)
 - Identify class-label attribute, if applicable
2. Find the central point for each attribute (mean, median, mode)
 - If they are very different → non-normal distribution, possible outliers
3. Understand the spread of each attribute (range, SD)
4. Watch out for outliers (boxplot, scatter plot)
 - Remove them
 - Choose methods that are robust to outliers
5. Visualize the relationship between attributes (scatter matrix)
6. Observe class patterns (parallel chart, deviation chart)